

# Michael F. Xu

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## Research Interest

My primary research interest is in the field of human-robot interaction, predominantly the social aspect of it, and integrating robotic technologies into human activities and settings. In particular, I am interested in long-term companion robot engagement and deployments, as well as their integration within family and school environments.

## Education

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|--------------|------------------------------------------------------------------------------------|----------------|
| <b>Ph.D.</b> | <b>University of Wisconsin–Madison</b> , Computer Sciences                         | 2023 – Present |
|              | • Advisor: Dr. Bilge Mutlu, Doctoral Minor: Human Ecology                          |                |
| <b>M.Sc.</b> | <b>Georgia Institute of Technology</b> , Computer Sciences                         | 2019 – 2021    |
| <b>M.A.</b>  | <b>Yale University</b> , Economics                                                 | 2014 – 2016    |
| <b>B.A.</b>  | <b>New York University</b> , Economics, and Mathematics ( <i>summa cum laude</i> ) | 2011 – 2014    |
| <b>B.A.</b>  | <b>Tsinghua University</b> , Economics and Finance                                 | 2010 – 2011    |

## Experience

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| <b>UW–Madison, Department of Computer Sciences</b> , Graduate Assistant                                                                                                                                                                                 | Madison, WI    |
| • Various research and teaching assistant roles within the department.                                                                                                                                                                                  | 2023 – Present |
| <b>Van Robotics</b> , Lead Software Architect                                                                                                                                                                                                           | Columbia, SC   |
| • Full Stack Engineer (2019 – 2020).                                                                                                                                                                                                                    | 2019 – 2023    |
| • Design and implement the high-level software architecture of the smart robot tutor ABii; Work with research team to perform data analysis, implement machine learning models, as well as deploying the models to edge devices in real-world settings. |                |
| <b>Music X Lab</b> , Research Intern                                                                                                                                                                                                                    | Remote         |
| • Advised by Dr. Gus G. Xia. Implemented hardware and software solutions for haptic music tutoring systems.                                                                                                                                             | 2022 – 2023    |
| <b>IQVIA</b> , Model Developer                                                                                                                                                                                                                          | New York, NY   |
| • Develop various Excel models; significant VBA codes used both to enhance user experience, and to compute model outputs (Epidemic/Monte Carlo models).                                                                                                 | 2017 – 2018    |

## Publications

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| <b>Xu, M. F.</b> , Zhao, E., Zhang, Y., Michaelis, J. E., Sebo, S., & Mutlu, B. (2026). <i>Designing Robots for Families: In-Situ Prototyping for Contextual Reminders on Family Routines</i> . In Proceedings of the 21st ACM/IEEE International Conference on Human-Robot Interaction (HRI) [23%] | 2026 |
| <b>Xu, M. F.</b> , Cagiltay, B., Hu, Y., Zhu, A., & Mutlu, B. (2026). <i>Designing Robots to Support Parent-Child Connections: Opportunities Through Robot-Mediated Communication</i> . In Proceedings of the 25th Interaction Design and Children Conference (IDC) [23%]                           |      |
| <b>Xu, M. F.</b> , Yang, Q., Kirkorian, H., & Mutlu, B. (2026). <i>Supporting Family-School Partnerships with Robot-Facilitated Home-Based Activities</i> . In Proceedings of the 25th Interaction Design and Children Conference (IDC) [23%]                                                       |      |
| <b>Xu, M. F.</b> , & Mutlu, B. (2025, March). <i>Exploring the Use of Robots for Diary Studies</i> . In 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI) [25%]                                                                                                          | 2025 |

He, X., **Xu, M. F.**, Cagiltay, B., & Mutlu, B. (2025, March). *Developing Robot Prototypes to Explore Robot-Facilitated Family Routines*. In 2025 20th ACM/IEEE International Conference on Human-Robot Interaction (HRI) [LBR]

**Xu, M. F.**, Cagiltay, B., Michaelis, J., Sebo, S., & Mutlu, B. (2024, August). *Robots in family routines: Development of and initial insights from the family-robot routines inventory*. In 2024 33rd IEEE International Conference on Robot and Human Interactive Communication (ROMAN) [Finalist: Pioneering Research Award]

2024

## Honors & Awards

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**Computer Sciences Summer Fellowship**, University of Wisconsin-Madison

2024

## Outreach

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### Grand Parents University

2025

- Served as the instructor for a two-day session, teaching grandparents and grandchildren how to program robots and interact with large-language models.

### UW-Madison Science Expeditions

- Hosted lab tours and stationed exhibition tables, performing multiple interactive robot demonstrations.

### Lab Tours for Schools

- Co-hosted many lab tours for classes of students from various grade levels (ranging from elementary to high school). Lab tours include a gentle introduction to the subject of robotics, and interactive demo sessions.

## Peer Review Service

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**Conferences:** ACM/IEEE HRI (2025–2026); ACM CHI (2025–2026); ACM IDC (2024, 2026)

**Journal:** ACM Transactions on Human-Robot Interaction (THRI)

## Personal Information

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**Languages:** Mandarin & English

**Citizenship:** U.S.A.